Purdue University 2019-2020 University Catalog

[ARCHIVED CATALOG]

First-Year Engineering

About the Program

The First-Year Engineering (FYE) Program, at the School of Engineering Education, is the entry point for all beginning engineering students at Purdue. In the FYE program, students complete foundational coursework in math, science, engineering, and communications (oral and written). After this common first year, students choose their discipline of engineering and start to follow the plan of study of a particular degree program. The mission of this student-oriented program is to advise, prepare, and retain outstanding students for degree programs in Purdue's College of Engineering.

Talk to an FYE Advisor

- make an appointment to meet with your FYE advisor
- learn about the walk-in schedule for FYE advisors or for Student Representatives of Professional Engineering Schools
- other common advisor questions

Requirements for First-Year Engineering (FYE)

30+ Credits Required

To complete the First-Year Engineering (FYE) program and be eligible for consideration for entry into an Engineering degree program, a student must comply with the grade average requirement and complete the following required courses:

Requirement #1: Intro to Engineering I (2-4 credits)

One of the following course options:

- ENGR 13100 - Transforming Ideas To Innovation I ♦ or
- ENGR 14100 - Honors Creativity And Innovation In Engineering Design I ♦ or
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I ♦
  OR
- EPCS 11100 - First Year Participation In EPICS I ♦ and
- EPCS 12100 - First Year Participation In EPICS II ♦

Requirement #2: Intro to Engineering II (2-4 credits)

One of the following course options:

- ENGR 13200 - Transforming Ideas To Innovation II ♦ or
- ENGR 13300 - Transforming Ideas To Innovation, EPICS ♦ or

- ENGR 14200 - Honors Creativity And Innovation In Engineering Design II ♦ or
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II ♦

Requirement #3: Calculus I (4-5 credits)

One of the following course options:

- MA 16100 - Plane Analytic Geometry And Calculus I ♦ *(satisfies Quantitative Reasoning for core)* or
- MA 16500 - Analytic Geometry And Calculus I ♦ *(satisfies Quantitative Reasoning for core)*

Requirement #4: Calculus II (4-5 credits)

One of the following course options:

- MA 16200 - Plane Analytic Geometry And Calculus II ♦ or
- MA 16600 - Analytic Geometry And Calculus II ♦

Requirement #5: Chemistry I (4-6 credits)

One of the following course options:

- CHM 11500 - General Chemistry ♦ *(satisfies Science for core)*
  OR
- CHM 11100 - General Chemistry ♦ *(satisfies Science for core)* and
- CHM 11200 - General Chemistry ♦ *(satisfies Science for core)*

Requirement #6: Physics (4 credits)

One of the following course options:

- PHYS 17200 - Modern Mechanics ♦ *(satisfies Science for core)*
  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I ♦ *(satisfies Science for core)* and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II ♦ *(satisfies Science for core)*

Requirement #7: Science Selective (3-4 credits)

One of the following course options:

- CHM 11600 - General Chemistry or ♦ *(satisfies Science for core)* or
- CS 15900 - C Programming or ♦ or
- BIOL 11000 - Fundamentals Of Biology I ♦ *(satisfies Science for core)* or
- BIOL 11100 - Fundamentals Of Biology II or ♦ *(satisfies Science for core)* or
- BIOL 13100 - Biology II: Development, Structure, And Function Of Organisms ♦ *(satisfies Science for core)*
  OR

- BIOL 12100 - Biology I: Diversity, Ecology, And Behavior ♦ *(satisfies Science for core)* and
- BIOL 13500 - First Year Biology Laboratory ♦ *(satisfies Science for core)*

Requirement #8: Written and Oral Communication (6 credits)

Two of the following course options:

- Written Communication - Credit Hours: 3.00-4.00 ♦ (satisfies Written Communication for core)
- Oral Communication - Credit Hours: 3.00 ♦ (satisfies Oral Communication for core)
- ENGL 11000 - American Language And Culture For International Students I ♦
- ENGL 11100 - American Language And Culture For International Students II ♦

Grade Average requirement

To complete the First-Year Engineering (FYE) program, the student's cumulative GPA must be of 2.00 or greater, and an Engineering Admissions Index (EAI) must be 2.00 or greater. Calculation of the EAI is equivalent to the calculation of GPA for courses used to meet all FYE requirements above. If a student meets a requirement in more than one way, only one will be used to calculate the EAI. The FYE Curriculum Committee will be responsible for keeping an updated, clear, and universal set of rules for determining which course is used in EAI for these situations. These rules are available to students in the FYE Advising office.

First Year Engineering Program Requirements

Fall 1st Year

- CHM 11500 - General Chemistry ♦ *(FYE Requirement #5)* - Credit Hours: 4.00
- ENGR 13100 - Transforming Ideas To Innovation I ♦ (*FYE Requirement #1)* - Credit Hours: 2.00
- MA 16100 - Plane Analytic Geometry And Calculus I ♦ *(FYE Requirement #3)* - Credit Hours: 5.00 or
- MA 16500 - Analytic Geometry And Calculus I ♦ *(FYE Requirement #3)* - Credit Hours: 4.00
- Written Communication Selective ♦ *(FYE Requirement #8)* - Credit Hours: 3.00-4.00 (Satisfies Written Communication for Core) or
- Oral Communication Selective ♦ *(FYE Requirement #8)* - Credit Hours:3.00 (Satisfies Oral Communication for Core)

13-14 Credits

Spring 1st Year

- ENGR 13200 - Transforming Ideas To Innovation II ♦ *(FYE Requirement #2)* - Credit Hours: 2.00
- PHYS 17200 - Modern Mechanics ♦ *(FYE Requirement #6)* - Credit Hours: 4.00
- MA 16200 - Plane Analytic Geometry And Calculus II ♦ (FYE Requirement #4) - Credit Hours: 5.00 or

- MA 16600 - Analytic Geometry And Calculus II ♦ *(FYE Requirement #4)* - Credit Hours: 4.00

- Oral Communication ♦ *(FYE Requirement #8)* - Credit Hours: 3.00 (Satisfies Oral Communication for core) or
- Written Communication ♦ *FYE Requirement #8)* - Credit Hours: 3.00-4.00 (Satisfies Written Communication for core)

- Science Selective ♦ *(FYE Requirement #7)* - Credit Hours: 3.00

16 Credits

Transitioning to a degree program

After completion of the First-Year Engineering (FYE) program, students may transition to one of fifteen degree programs in engineering (i.e., Mechanical, Electrical, Chemical, Civil, etc.); this happens through the "Transition to Major" (or T2M) process. As a student nears completion of FYE, he or she will inform FYE of their preferred choices of degree program (major). If the degree program is not at capacity, all students completing FYE requirements and requesting that degree program will be admitted. If the program is at capacity, admission will be based on academic record, including GPA, EAI, and grades in specific courses.

Note

- All courses used to fulfill the First-Year Engineering program requirements must have a grade of C- or higher

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements.

The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.